



C23-C-405

23428

BOARD DIPLOMA EXAMINATION, (C-23)
MARCH/APRIL—2025
DCE – FOURTH SEMESTER EXAMINATION
IRRIGATION ENGINEERING

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Distinguish between perennial irrigation and inundation irrigation.
2. Write the Ryve's and Dicken's formulae.
3. Distinguish between weir and barrage.
4. Define the terms gravity dam and free board.
5. List the various forces acting on a gravity dam.
6. Write short note on 'Berm'.
7. State the need for canal lining.
8. Define river training works.
9. List different types of groynes.
10. Write the advantages of sprinkler irrigation.

PART—B

10×5=50

- Instructions :** (1) Answer *any* **five** questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. Explain the factors affecting the run-off.
12. Sketch and explain lay out of diversion head works.
13. Describe the causes of failures in gravity dams.
14. Explain the causes of failures in earth dams.
15. Explain the different methods of alignment of canals with a neat sketch.
16. Explain the aqueduct and super passage with a neat sketch.
17. Explain the methods of preventing soil erosion.
18. Explain about different stages of flow of rivers.

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